



KTU NOTES

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Module 5

Solution of difference equations using Z-T

z-transfer fn., $H(z) = \frac{C(z)}{R(z)} = \frac{O/P}{I/P}$

→ Find the impulse response and step

response for the following system

a) $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$

b) $y(n) = x(n) + 2x(n-1) - 4x(n-2) + x(n-3)$

a: a) $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n)$

Taking z.T OBS

$$Y(z) - \frac{3}{4}[z^{-1}Y(z) + y(-1)] + \frac{1}{8}[z^{-2}Y(z) + z^{-1}y(-1) + y(-2)] = X(z)$$

$$y(-1) = y(-2) = 0$$

$$Y(z) - \frac{3}{4} z^{-1} Y(z) + \frac{1}{8} z^{-2} Y(z) = X(z)$$

$$Y(z) \left[1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2} \right] = X(z)$$

$$ZTF = \frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}}$$

Impulse response, $x(t) = \delta(t)$
 $X(z) = 1$

$$Y(z) = \frac{1}{1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}}$$

$$= \frac{z^2}{z^2 - \frac{3}{4} z + \frac{1}{8}}$$

$$= \frac{\cancel{8z^2}}{\cancel{8z^2} - 6z + 1}$$

$$\begin{aligned} s_1 &= \frac{3}{4} \\ p_2 &= \frac{1}{8} \end{aligned} \quad \left\{ \begin{array}{l} \frac{1}{2} \\ \frac{1}{4} \end{array} \right.$$

$$\frac{Y(z)}{z} = \frac{z}{z^2 - \frac{3}{4} z + \frac{1}{8}}$$

$$= \frac{z}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}$$

$$= \frac{A}{z - 1/2} + \frac{B}{z - 1/4}$$

$$\begin{aligned}
 A &= \left(\cancel{z - \frac{1}{2}} \right) \frac{z}{(\cancel{z - \frac{1}{2}})(z - \frac{1}{4})} \Big|_{z = \frac{1}{2}} \\
 &= \frac{\frac{1}{2}}{\frac{1}{2} - \frac{1}{4}} = \frac{\frac{1}{2}}{\frac{1-2}{1 \times 2}} = \frac{\frac{1}{2}}{\frac{2}{8}} = \frac{1}{2} \times \frac{8}{2} \\
 &= \underline{\underline{2}}
 \end{aligned}$$

$$\begin{aligned}
 B &= \left(\cancel{z - \frac{1}{4}} \right) \frac{z}{(z - \frac{1}{2})(\cancel{z - \frac{1}{4}})} \Big|_{z = \frac{1}{4}} \\
 &= \frac{\frac{1}{4}}{\frac{1}{4} - \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{2-4}{4 \times 2}} = \frac{\frac{1}{4}}{\frac{-2}{8}} = \frac{1}{4} \times \frac{-8}{2} \\
 &= \underline{\underline{-1}}
 \end{aligned}$$

$$\frac{Y(z)}{z} = \frac{2}{z - \frac{1}{2}} - \frac{1}{z - \frac{1}{4}}$$

$$Y(z) = \frac{2z}{z - \frac{1}{2}} - \frac{z}{z - \frac{1}{4}}$$

$$y(n) = 2 \left(\frac{1}{2} \right)^n u(n) - \left(\frac{1}{4} \right)^n u(n)$$

Step response

$$x(n) = u(n)$$

$$X(z) = \frac{z}{z-1}$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

$$Y(z) = \frac{z}{z-1} \times \frac{z^2}{z^2 - \frac{3}{4}z + \frac{1}{8}}$$

$$\frac{Y(z)}{z} = \frac{z^2}{(z-1)\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}$$

$$\frac{Y(z)}{z} = \frac{A}{z-1} + \frac{B}{z-\frac{1}{2}} + \frac{C}{z-\frac{1}{4}}$$

$$A = \left. (z-1) \frac{z^2}{(z-1)\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)} \right|_{z=1} = \frac{1}{\left(1-\frac{1}{2}\right)\left(1-\frac{1}{4}\right)}$$

$$= \frac{1}{\frac{1}{2} \times \frac{3}{4}} = \underline{\underline{\frac{8}{3}}}$$

$$\begin{aligned}
 B &= \left(\cancel{z - \frac{1}{2}} \right) \frac{z^2}{(z-1)(\cancel{z - \frac{1}{2}})(z - \frac{1}{4})} \Big|_{z = \frac{1}{2}} \\
 &= \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{1}{2} - 1\right) \left(\frac{1}{2} - \frac{1}{4}\right)} = \frac{\frac{1}{4}}{\left(-\frac{1}{2}\right) \cdot \frac{1}{4}} \\
 &= \underline{\underline{-2}}
 \end{aligned}$$

$$\begin{aligned}
 C &= \left(\cancel{z - \frac{1}{4}} \right) \frac{z^2}{(z-1)(z - \frac{1}{2})(\cancel{z - \frac{1}{4}})} \Big|_{z = \frac{1}{4}} \\
 &= \frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{4} - 1\right) \left(\frac{1}{4} - \frac{1}{2}\right)} = \frac{\frac{1}{16}}{\frac{-3}{4} \times \frac{-1}{4}} \\
 &= \frac{1}{16} \times \frac{16}{3} = \underline{\underline{\frac{1}{3}}}
 \end{aligned}$$

$$\frac{Y(z)}{z} = \frac{8}{3} \frac{1}{z-1} + \frac{-2}{z - \frac{1}{2}} + \frac{1}{3} \frac{1}{z - \frac{1}{4}}$$

$$Y(z) = \frac{8}{3} \frac{z}{z-1} - \frac{2z}{z - \frac{1}{2}} + \frac{1}{3} \frac{z}{z - \frac{1}{4}}$$

$$y(n) = \frac{8}{3} u(n) - 2 \left(\frac{1}{2}\right)^n u(n) + \frac{1}{3} \left(\frac{1}{4}\right)^n u(n)$$

$$b) y(n) = x(n) + 2x(n-1) - 4x(n-2) + x(n-3)$$

$$Y(z) = X(z) + 2z^{-1}X(z) - 4z^{-2}X(z) + z^{-3}X(z)$$

$$\frac{Y(z)}{X(z)} = 1 + 2z^{-1} - 4z^{-2} + z^{-3}$$

Impulse response $x(n] = \delta(n)$
 $X(z) = 1$

$$Y(z) = 1 + 2z^{-1} - 4z^{-2} + z^{-3}$$

$$y(n) = \{1, 2, -4, 1\}$$

↑

Step response $x(n) = u(n) ; X(z) = \frac{z}{z-1}$

$$Y(z) = \frac{z}{z-1} [1 + 2z^{-1} - 4z^{-2} + z^{-3}]$$

$$= \frac{1}{1-z^{-1}} [1 + 2z^{-1} - 4z^{-2} + z^{-3}]$$

$$= \frac{1}{1-z^{-1}} + \frac{2z^{-1}}{1-z^{-1}} - 4 \frac{z^{-2}}{1-z^{-1}} + \frac{z^{-3}}{1-z^{-1}}$$

Using time shifting prop.

$$y(n) = u(n) + 2u(n-1) - 4u(n-2) + u(n-3)$$

→ Find the output $y(n)$ of a LTI discrete time s/m specified by the equation.

$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = 2x(n) + \frac{3}{2}x(n-1)$$

if the initial condition are $y(-1) = 0, y(-2) = 1$

and the input $x(n] = \left(\frac{1}{4}\right)^n u(n)$.

A:
$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = 2x(n) + \frac{3}{2}x(n-1)$$

Taking Z.T.

$$\begin{aligned} Y(z) - \frac{3}{2}[z^{-1}Y(z) + y(-1)] + \frac{1}{2}[z^{-2}Y(z) + z^{-1}y(-1) + y(-2)] \\ = 2X(z) + \frac{3}{2}[z^{-1}X(z) + x(-1)] \end{aligned}$$

Given $y(-1) = 0 ; y(-2) = 1 ; x(-1) = 0$

$$Y(z) - \frac{3}{2} z^{-1} Y(z) + \frac{1}{2} [z^{-2} Y(z) + 1] = 2X(z) + \frac{3}{2} z^{-1} X(z)$$

$$Y(z) \left[1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2} \right] + \frac{1}{2} = \left[2 + \frac{3}{2} z^{-1} \right] X(z)$$

Given $x(n) = \left(\frac{1}{4}\right)^n u(n)$

$$X(z) = \frac{z}{z - \frac{1}{4}} = \frac{1}{1 - \frac{1}{4} z^{-1}}$$

$$Y(z) \left[1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2} \right] + \frac{1}{2} = \left(2 + \frac{3}{2} z^{-1} \right) \times \frac{1}{1 - \frac{1}{4} z^{-1}}$$

$$Y(z) \left[1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2} \right] = \frac{-1}{2} + \frac{2 + \frac{3}{2} z^{-1}}{1 - \frac{1}{4} z^{-1}}$$

$$Y(z) = \frac{-1}{2 \left(1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2} \right)} + \frac{2 + \frac{3}{2} z^{-1}}{\left(1 - \frac{1}{4} z^{-1} \right) \left(1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2} \right)}$$

$$Y(z) = \frac{-z^2}{2 \left(z^2 - \frac{3}{2} z + \frac{1}{2} \right)} + \frac{z^3 \left(2 + \frac{3}{2} z^{-1} \right)}{\left(z - \frac{1}{4} \right) \left(z^2 - \frac{3}{2} z + \frac{1}{2} \right)}$$

$$= \frac{-z^2}{2 \left(z^2 - \frac{3}{2} z + \frac{1}{2} \right)} + \frac{z^2 \left(2z + \frac{3}{2} \right)}{\left(z - \frac{1}{4} \right) \left(z^2 - \frac{3}{2} z + \frac{1}{2} \right)}$$

$$= \frac{-z^2}{2(z-1)(z-\frac{1}{2})} + \frac{z^2(2z+\frac{3}{2})}{(z-\frac{1}{4})(z-1)(z-\frac{1}{2})}$$

$$= Y_1(z) + Y_2(z).$$

$$Y_1(z) = \frac{-z^2}{2(z-1)(z-\frac{1}{2})} \Rightarrow \frac{A}{z}$$

$$\frac{Y_1(z)}{z} = \frac{-z}{2(z-1)(z-\frac{1}{2})} = \frac{A}{z-1} + \frac{B}{z-\frac{1}{2}}$$

$$A = \cancel{(z-1)} \cdot \frac{-z}{2\cancel{(z-1)}(z-\frac{1}{2})} \Big|_{z=1} = \frac{-1}{2(1-\frac{1}{2})} = \underline{\underline{-1}}$$

$$B = \cancel{(z-\frac{1}{2})} \cdot \frac{-z}{2(z-1)\cancel{(z-\frac{1}{2})}} \Big|_{z=\frac{1}{2}} = \frac{-\frac{1}{2}}{2(\frac{1}{2}-1)} = \frac{-\frac{1}{2}}{2 \times -\frac{1}{2}} = \underline{\underline{\frac{1}{2}}}$$

$$\frac{Y_1(z)}{z} = \frac{-1}{z-1} + \frac{\frac{1}{2}}{z-\frac{1}{2}}$$

$$Y_1(z) = \frac{-z}{z-1} + \frac{\frac{1}{2}z}{z-\frac{1}{2}} \Rightarrow y_1(n) = -u(n) + \frac{1}{2} \left(\frac{1}{2}\right)^n u(n)$$

$$Y_2(Z) = \frac{Z^2 (2Z + \frac{3}{2})}{(Z - \frac{1}{4})(Z - 1)(Z - \frac{1}{2})}$$

$$\frac{Y_2(Z)}{Z} = \frac{Z (2Z + \frac{3}{2})}{(Z - \frac{1}{4})(Z - 1)(Z - \frac{1}{2})} = \frac{C}{Z - \frac{1}{4}} + \frac{D}{Z - 1} + \frac{E}{Z - \frac{1}{2}}$$

$$C = \left(\cancel{Z - \frac{1}{4}} \right) \cdot \frac{Z (2Z + \frac{3}{2})}{(\cancel{Z - \frac{1}{4}})(Z - 1)(Z - \frac{1}{2})} \Big|_{Z = \frac{1}{4}}$$

$$= \frac{\frac{1}{4} \left(2 \times \frac{1}{4} + \frac{3}{2} \right)}{\left(\frac{1}{4} - 1 \right) \left(\frac{1}{4} - \frac{1}{2} \right)} = \frac{\frac{1}{4} \times \frac{7}{2}}{\frac{-3}{4} \times \frac{-1}{4}} = \frac{\frac{7}{8}}{\frac{3}{16}}$$

$$= \underline{\underline{\frac{8}{3}}}$$

$$D = \frac{(\cancel{Z - 1}) \cdot Z (2Z + \frac{3}{2})}{(Z - \frac{1}{4})(\cancel{Z - 1})(Z - \frac{1}{2})} \Big|_{Z = 1} = \frac{2 + \frac{3}{2}}{(1 - \frac{1}{4})(1 - \frac{1}{2})}$$

$$= \frac{\frac{7}{2}}{\frac{3}{4} \times \frac{1}{2}} = \frac{7}{2} \times \frac{8}{3} = \frac{28}{3}$$

$$E = \left(\cancel{Z - \frac{1}{2}} \right) \cdot \frac{Z (2Z + \frac{3}{2})}{(Z - \frac{1}{4})(Z - 1)(\cancel{Z - \frac{1}{2}})} \Big|_{Z = \frac{1}{2}} = \frac{\frac{1}{2} \left(1 + \frac{3}{2} \right)}{\left(\frac{1}{2} - \frac{1}{4} \right) \left(\frac{1}{2} - 1 \right)}$$

$$= \frac{\frac{1}{2} \times \frac{5}{2}}{\frac{1}{4} \times -\frac{1}{2}} = \underline{\underline{-10}}$$

$$\frac{Y_2(z)}{z} = \frac{8}{3} \frac{1}{z-1/4} + \frac{28}{3} \frac{1}{z-1} + \frac{-10}{z-1/2}$$

$$Y_2(z) = \frac{8}{3} \frac{z}{z-1/4} + \frac{28}{3} \frac{z}{z-1} - \frac{10z}{z-1/2}$$

$$y_2(n) = \frac{8}{3} \left(\frac{1}{4}\right)^n u(n) + \frac{28}{3} u(n) - 10 \left(\frac{1}{2}\right)^n u(n)$$

$$y(n) = y_1(n) + y_2(n)$$

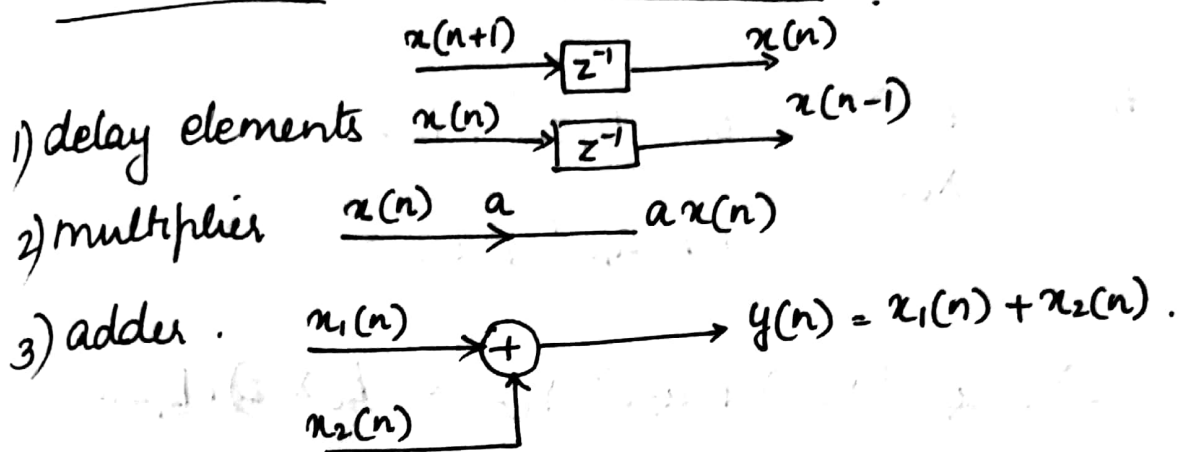
$$= -u(n) + \frac{1}{2} \cdot \left(\frac{1}{2}\right)^n u(n) +$$

$$\frac{8}{3} \cdot \left(\frac{1}{4}\right)^n u(n) + \frac{28}{3} u(n) - 10 \left(\frac{1}{2}\right)^n u(n)$$

$$= \frac{25}{3} u(n) - \frac{19}{2} \cdot \left(\frac{1}{2}\right)^n u(n) + \frac{8}{3} \cdot \left(\frac{1}{4}\right)^n u(n)$$

$\frac{28}{3} - 10 \cdot \frac{1}{2}$
 $\frac{1}{2} - 10$
 $\frac{1-20}{2}$

Block diagram representation



4 types of realizations

1. Direct form-I
2. Direct form-II
3. Cascade form
4. Parallel form.

→ Obtain the direct form I realization.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

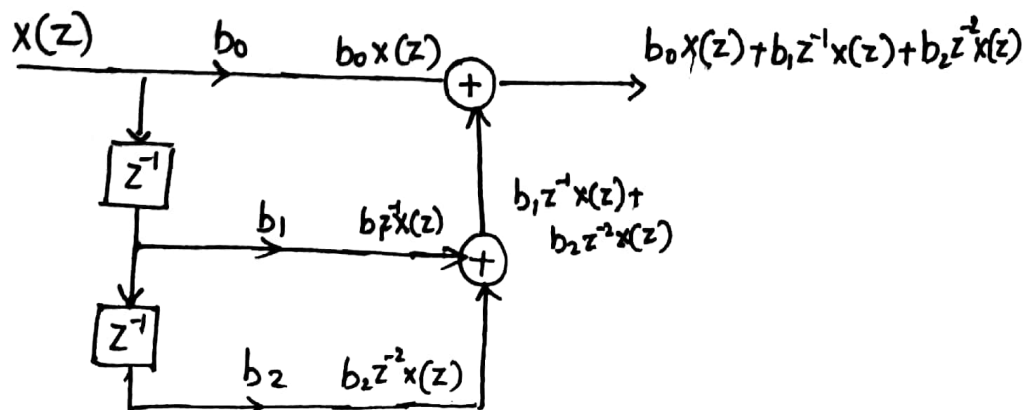
$$Y(z) + a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) = b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z)$$

$$\text{Let } b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z) = W(z) \quad \text{--- (1)}$$

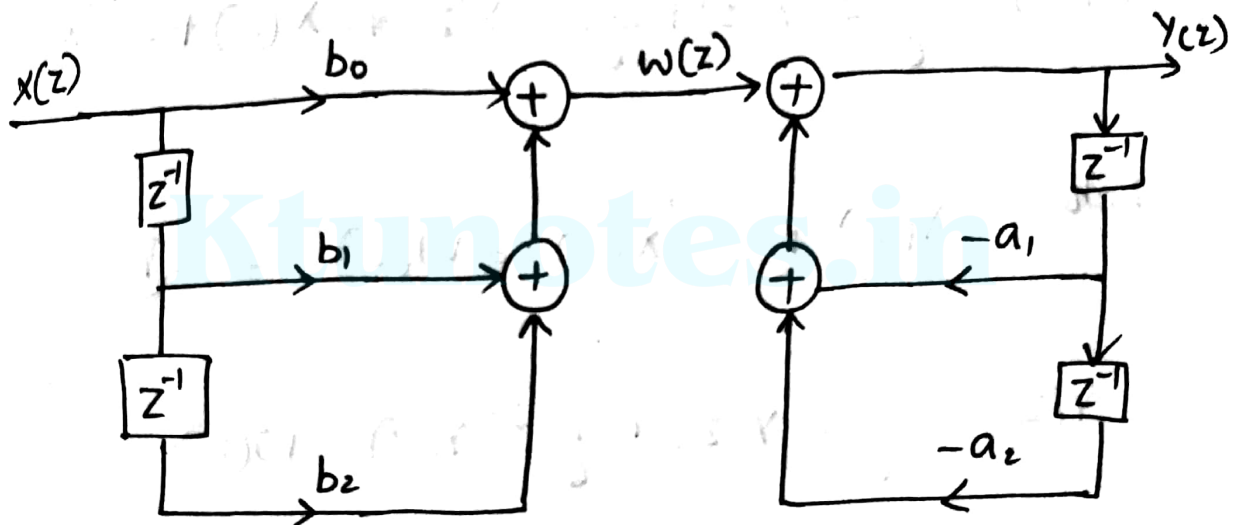
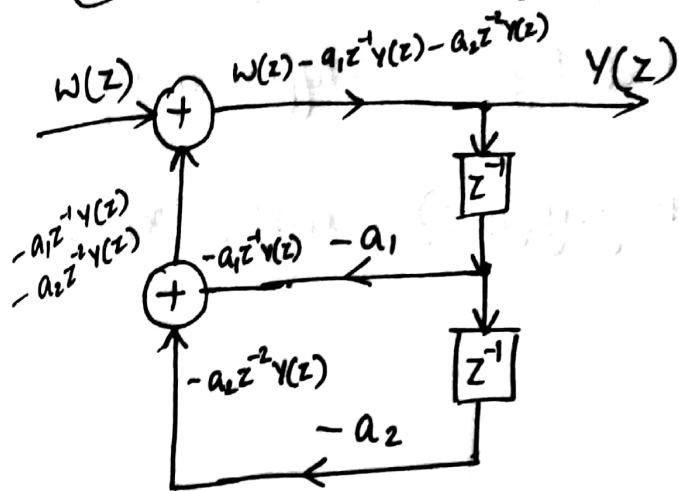
$$\therefore Y(z) + a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) = W(z)$$

$$Y(z) = W(z) - a_1 z^{-1} Y(z) - a_2 z^{-2} Y(z) \quad \text{--- (2)}$$

① can be realized as.



② can be realized as .



Direct I realization.

→ Obtain the direct form-II realization for the system described by the difference eq.

$$y(n) - \frac{5}{6} y(n-1] + \frac{1}{6} y(n-2) = x(n) + 2x(n-1]$$

A: Taking z.T.

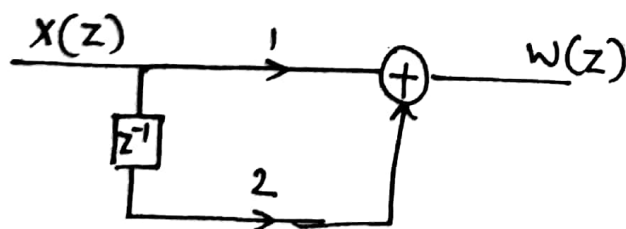
$$Y(z) - \frac{5}{6} z^{-1} Y(z) + \frac{1}{6} z^{-2} Y(z) = X(z) + 2z^{-1} X(z)$$

Let $X(z) + 2z^{-1} X(z) = W(z)$ ——— ①

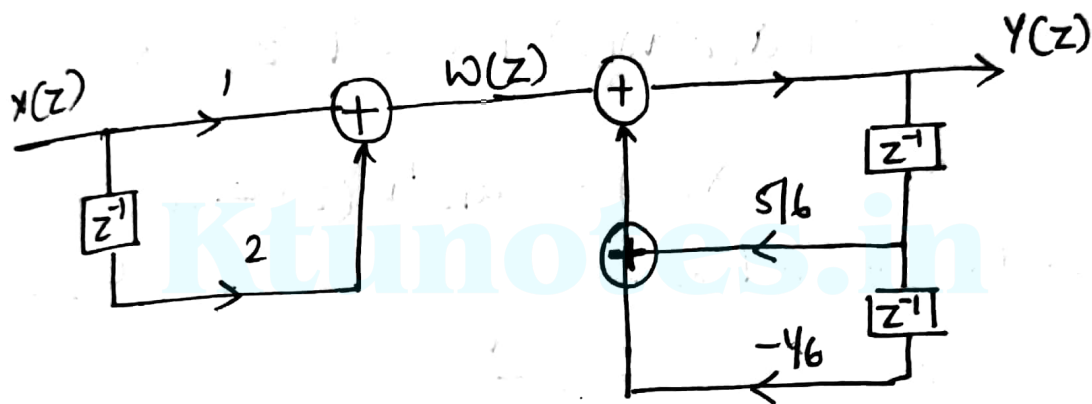
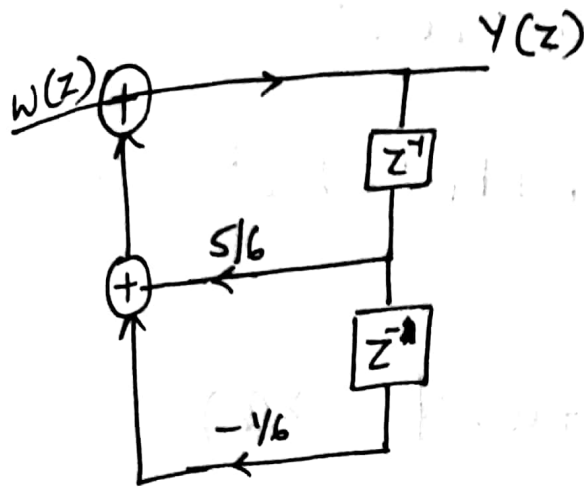
$$Y(z) - \frac{5}{6} z^{-1} Y(z) + \frac{1}{6} z^{-2} Y(z) = W(z)$$

$$Y(z) = W(z) + \frac{5}{6} z^{-1} Y(z) - \frac{1}{6} z^{-2} Y(z) \rightarrow ②$$

① can be realised as .



② can be realized as .



Direct form II realization

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_1 + a_1 z^{-1} + a_2 z^{-2}}$$

$$\frac{Y(z)}{W(z)} \times \frac{W(z)}{X(z)}$$

where $\frac{w(z)}{x(z)} = \frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}} \quad \text{--- (1)}$

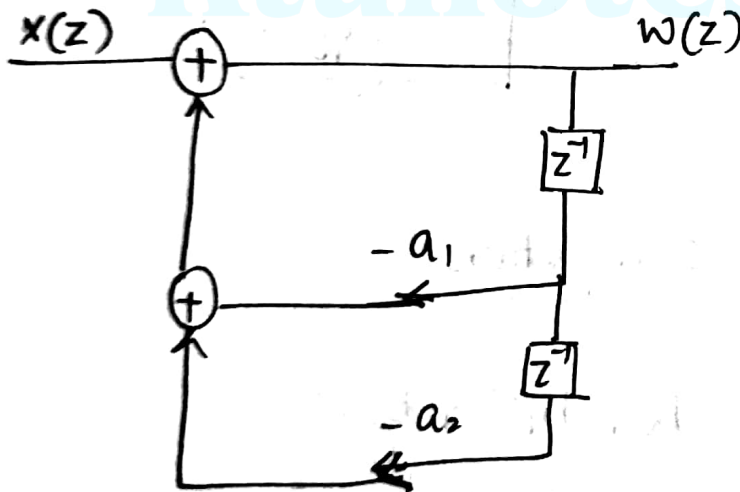
$$\frac{y(z)}{w(z)} = b_0 + b_1 z^{-1} + b_2 z^{-2} \quad \text{--- (2)}$$

$$w(z) [1 + a_1 z^{-1} + a_2 z^{-2}] = x(z)$$

$$w(z) + a_1 z^{-1} w(z) + a_2 z^{-2} w(z) = x(z)$$

$$w(z) = x(z) - a_1 z^{-1} w(z) - a_2 z^{-2} w(z)$$

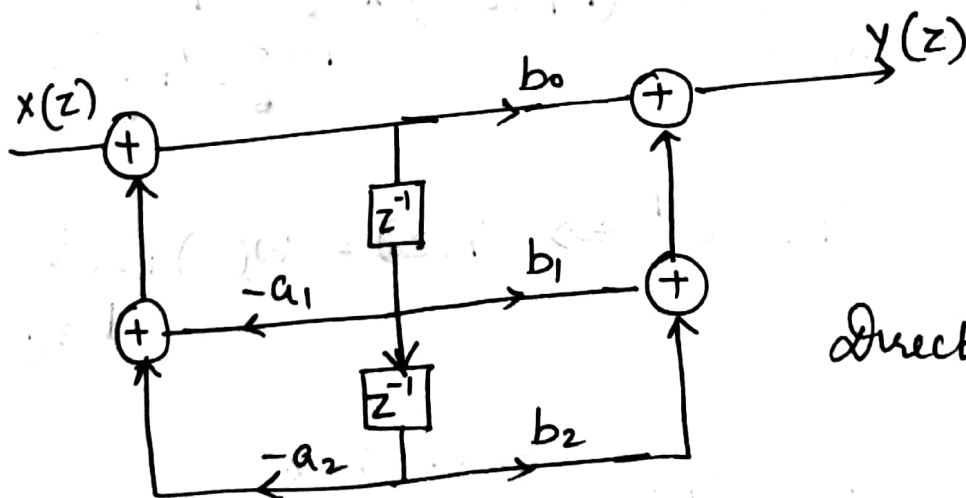
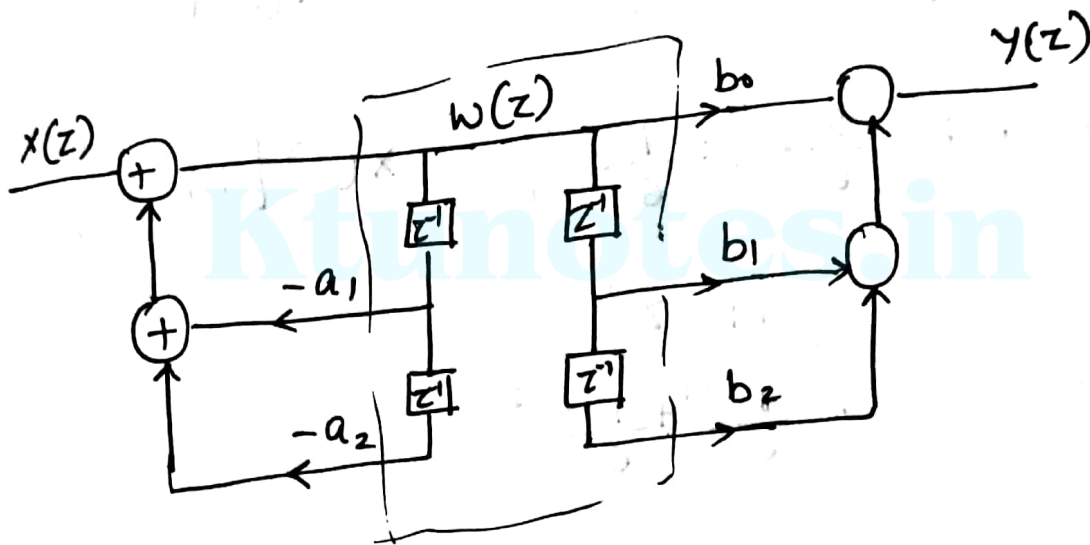
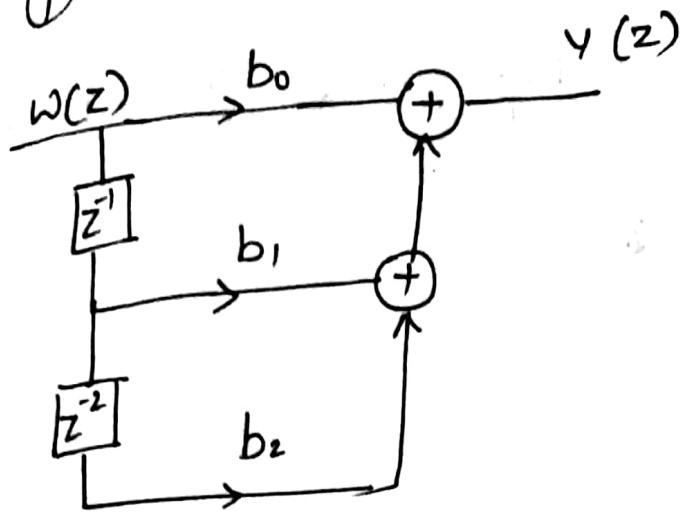
③ can be realised as ! ↳ (3)



$$\textcircled{2} \Rightarrow y(z) = b_0 w(z) + b_1 z^{-1} w(z) + b_2 z^{-2} w(z)$$

↳ (4)

④ can be realized as .



Direct form - II

→ Realize the system

$$y(n) - \frac{5}{6} y(n-1) + \frac{1}{6} y(n-2) = x(n) + 2x(n-1)$$

in direct form - II.

A: Taking ZT.

$$Y(z) - \frac{5}{6} z^{-1} Y(z) + \frac{1}{6} z^{-2} Y(z) = X(z) + 2z^{-1} X(z)$$

$$Y(z) \left[1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2} \right] = X(z) [1 + 2z^{-1}]$$

$$\frac{Y(z)}{X(z)} = \frac{1 + 2z^{-1}}{1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2}}$$

$$\frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \times \frac{W(z)}{X(z)} = (1 + 2z^{-1}) \times \frac{1}{1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2}}$$

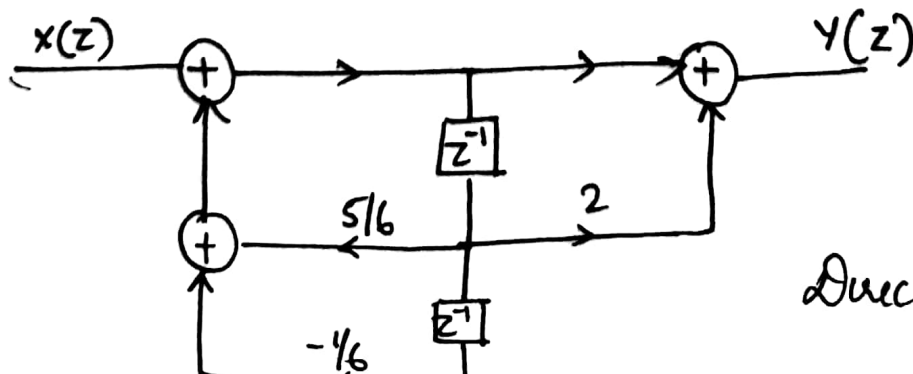
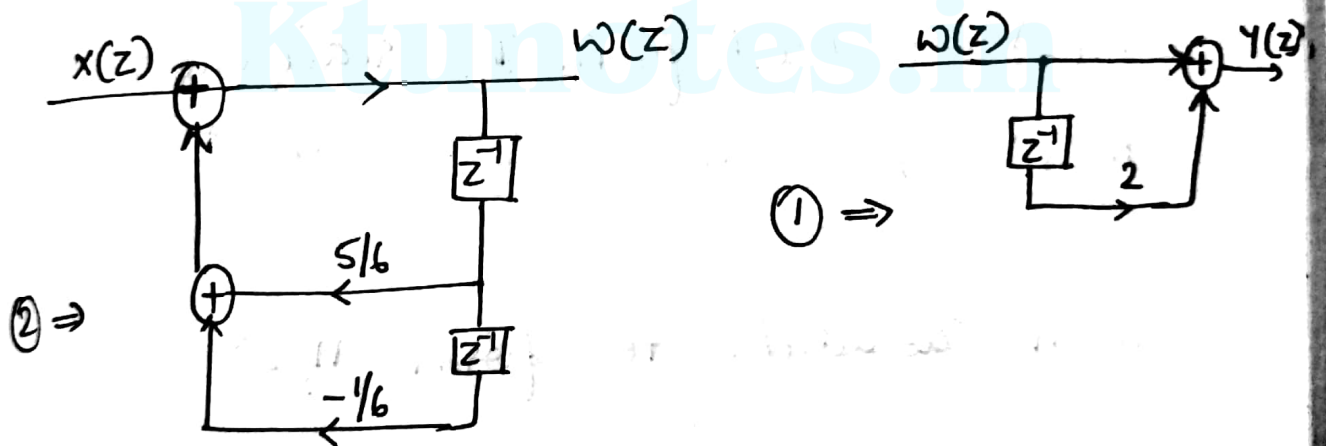
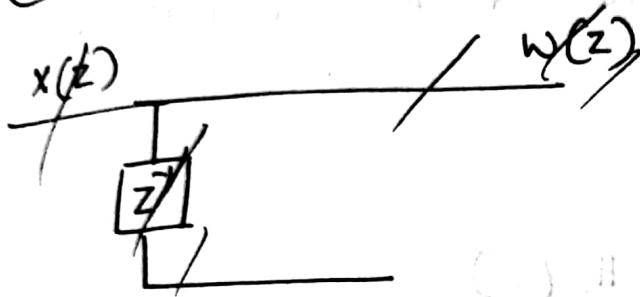
$$\frac{Y(z)}{W(z)} = 1 + 2z^{-1} \Rightarrow Y(z) = W(z) + 2z^{-1} W(z) \quad \hookrightarrow \textcircled{1}$$

$$\frac{W(z)}{X(z)} = \frac{1}{1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2}} \Rightarrow W(z) = X(z)$$

$$W(z) \left[1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2} \right] = X(z)$$

$$W(z) = X(z) + \frac{5}{6} z^{-1} W(z) - \frac{1}{6} z^{-2} W(z) \quad \text{②}$$

② can be realized as .



Direct form II

Cascade form

Given s/m function is expressed as a product of several system functions. Each of system fn. is realised in direct form II and cascaded.

$$H(z) = H_1(z) \cdot H_2(z).$$

$H_1(z) \rightarrow$ direct form II realization

$H_2(z) \rightarrow$ " " "

then cascaded to form $H(z)$.

-1/2 1/4
✓

→ Obtain the cascade form realisation of the s/m described by the difference equation.

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + 3x(n-1) + 2x(n-2)$$

Ans: Taking ZT

$$Y(z) - \frac{1}{4}Z^{-1}Y(z) - \frac{1}{8}Z^{-2}Y(z) = X(z) + 3Z^{-1}X(z) + 2Z^{-2}X(z)$$

$$Y(z) \left[1 - \frac{1}{4}Z^{-1} - \frac{1}{8}Z^{-2} \right] = X(z) [1 + 3Z^{-1} + 2Z^{-2}]$$

$$\frac{Y(z)}{X(z)} = \frac{1 + 3Z^{-1} + 2Z^{-2}}{1 - \frac{1}{4}Z^{-1} - \frac{1}{8}Z^{-2}} = \frac{(1+Z^{-1})(1+2Z^{-1})}{\left(1 - \frac{1}{2}Z^{-1}\right)\left(1 + \frac{1}{4}Z^{-1}\right)}$$

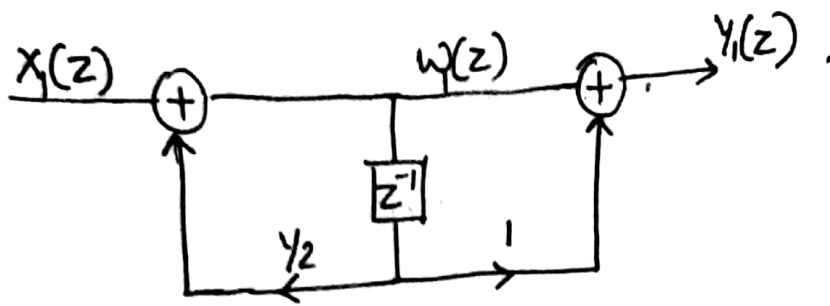
$$= H_1(z) \cdot H_2(z)$$

$$\left\{ \begin{array}{l} z = \frac{1}{2} \\ p = -\frac{1}{4} \end{array} \right\} \begin{array}{l} -\frac{1}{2} \\ +\frac{1}{4} \end{array}$$

$$H_1(z) = \frac{1+Z^{-1}}{1 - \frac{1}{2}Z^{-1}} ; H_2(z) = \frac{1+2Z^{-1}}{1 + \frac{1}{4}Z^{-1}}$$

$$\cancel{X(z)} \text{ (1)} = \frac{Y_1(z)}{W_1(z)} \times \frac{W_1(z)}{X_1(z)}$$

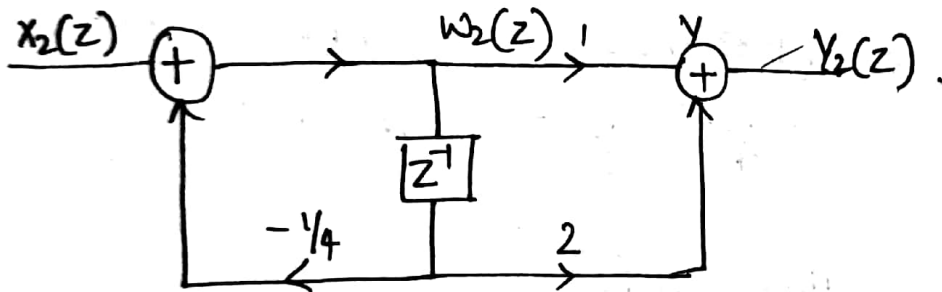
$$\frac{Y(z)}{W_1(z)} = 1+Z^{-1} ; \frac{W_1(z)}{X_1(z)} = \frac{1}{1 - \frac{1}{2}Z^{-1}} \Rightarrow W_1(z) = X_1(z) + \frac{1}{2}Z^{-1}W_1(z)$$



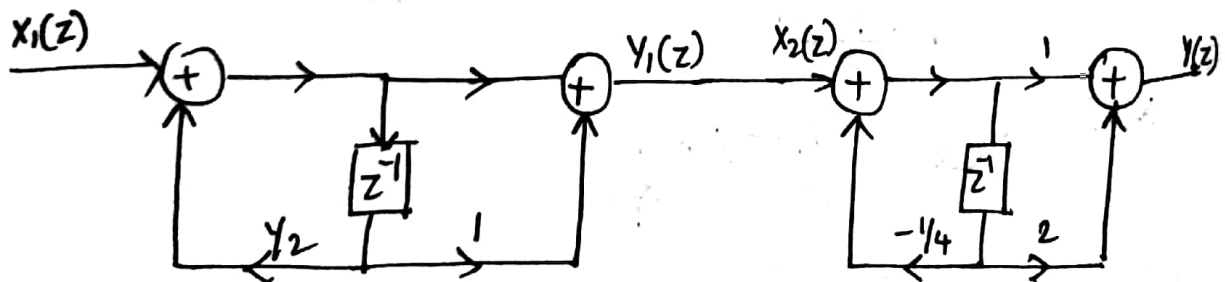
$$H_2(z) = \frac{1+2z^{-1}}{1+\frac{1}{4}z^{-1}} = \frac{Y_2(z)}{X_2(z)} = \frac{Y_2(z)}{W_2(z)} \times \frac{W_2(z)}{X_2(z)}$$

$$\frac{Y_2(z)}{W_2(z)} = 1+2z^{-1} \quad ; \quad \frac{W_2(z)}{X_2(z)} = \frac{1}{1+\frac{1}{4}z^{-1}}$$

$$W_2(z) = X_2(z) - \frac{1}{4}z^{-1}W_2(z)$$



Cascade two direct II forms.



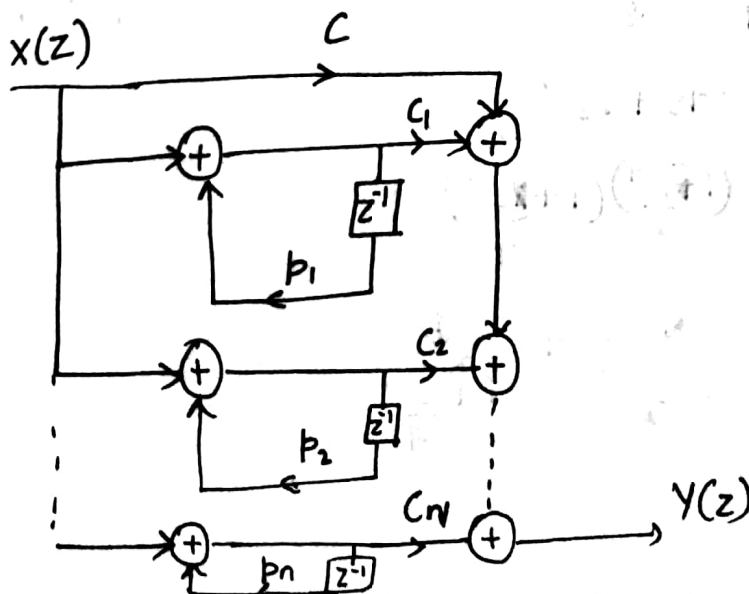
Parallel form realization

Parallel form realization of a system can be obtained by expressing the system function $H(s)$ in partial fraction as

$$H(s) = C + \sum_{k=1}^N \frac{C_k}{1 - p_k z^{-1}} \quad p_k \rightarrow \text{poles}$$

$$= C + \frac{C_1}{1 - p_1 z^{-1}} + \frac{C_2}{1 - p_2 z^{-1}} + \dots + \frac{C_N}{1 - p_N z^{-1}}$$

Each factor in the system is realized in direct form II and connected in parallel.



→ Obtain parallel form realisation of the discrete time system

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + 3x(n-1) + 2x(n-2)$$

A: Taking Z.T

$$Y(z) - \frac{1}{4}z^{-1}Y(z) - \frac{1}{8}z^{-2}Y(z) = X(z) + 3z^{-1}X(z) + 2z^{-2}X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{1 + 3z^{-1} + 2z^{-2}}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \quad -\frac{1}{8} \times 16$$

$$\frac{\frac{1}{8}z^{-2} - \frac{1}{4}z^{-1} + 1}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \left| \frac{1 + 3z^{-1} + 2z^{-2}}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \right.$$

$$\frac{Y(z)}{X(z)} = 16 + \frac{7z^{-1} - 15}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

$$-\frac{1}{8}z^{-2} - \frac{1}{4}z^{-1} + 1 \left| \frac{2z^{-2} + 3z^{-1} + 1}{2z^{-2} - 4z^{-1} + 16} \right. \\ \frac{7z^{-1} - 15}{7z^{-1} - 15}$$

$$= 16 + \frac{-15 + 7z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{1}{4}z^{-1}\right)}$$

$$= 16 + \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 + \frac{1}{4}z^{-1}}$$

$$= 16 + \frac{z^2}{z^2} \frac{-15+7z^{-1}}{(1-\frac{1}{2}z^{-1})(1+\frac{1}{4}z^{-1})}$$

$$H(z) = 16 + \frac{z(7-15z)}{(z-\frac{1}{2})(z+\frac{1}{4})}$$

$$\frac{H(z)}{z} = \frac{16}{z} + \frac{7-15z}{(z-\frac{1}{2})(z+\frac{1}{4})}$$

$$\frac{7-15z}{(z-\frac{1}{2})(z+\frac{1}{4})} = \frac{A}{z-\frac{1}{2}} + \frac{B}{z+\frac{1}{4}}$$

$$A = \left. \frac{7-15z}{z+\frac{1}{4}} \right|_{z=\frac{1}{2}} = \frac{7-15/2}{\frac{1}{2}+\frac{1}{4}} = \frac{-1/2}{6/8}$$

$$= \frac{-1}{2} \times \frac{8}{6} = \frac{-2}{3}$$

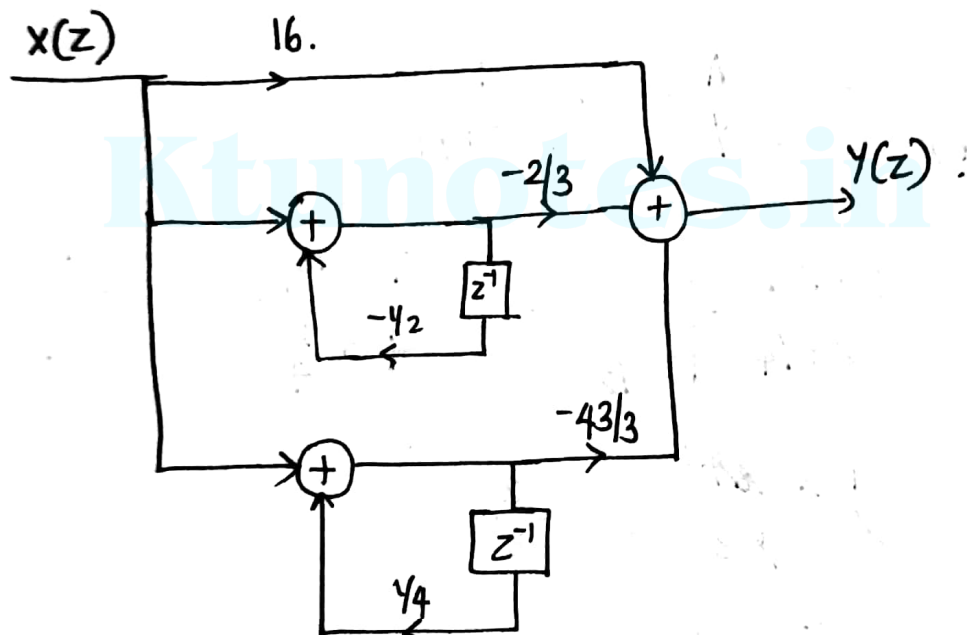
$$B = \left. \frac{7-15z}{z-\frac{1}{2}} \right|_{z=-1/4} = \frac{7+15/4}{-\frac{1}{4}-\frac{1}{2}} = \frac{43/4}{-6/8}$$

$$= \frac{43}{4} \times \frac{8}{-6} = \frac{-43}{3}$$

$$\frac{H(z)}{z} = \frac{16}{z} + \frac{-2}{3} \cdot \frac{1}{z - \frac{1}{2}} - \frac{43}{3} \frac{1}{z + \frac{1}{4}}$$

$$H(z) = 16 - \frac{2}{3} \frac{z}{z - \frac{1}{2}} - \frac{43}{3} \frac{z}{z + \frac{1}{4}}$$

$$H(z) = 16 - \frac{2}{3} \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{43}{3} \frac{1}{1 + \frac{1}{4}z^{-1}}$$



Stability of Sampled data Systems

The sampled data control s/m is stable if all the poles of z-transfer function of the system lies inside the unit circle in z-plane

$$\text{Z.TF}, H(z) = \frac{C(z)}{R(z)}$$

Methods for stability analysis

1. Jury's stability test.
2. Bilinear Transformation

Jury's Stability Test

is used to determine whether the roots of the characteristic polynomial lie within a unit circle or not.

Consist of two parts .

Necessary condition for stability

Sufficient condition for stability

n^{th} order characteristic polynomial $F(z)$ is

$$F(z) = a_n z^n + a_{n-1} z^{n-1} + a_{n-2} z^{n-2} + \dots + a_1 z + a_0 = 0$$

The necessary cond. to be satisfied for the stability of the s/m are

$$F(1) > 0 \quad \text{and} \quad (-1)^n F(-1) > 0.$$

The sufficient condition for stability

Row	z^0	z^1	z^2	$\dots$	z^n	$F(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0$
1	a_0	a_1	a_2	$\dots$	a_n	$\rightarrow$ coeff.
2	a_n	a_{n-1}	a_{n-2}	$\dots$	a_1	$\rightarrow$ reverse order of 1st row.
3	b_0	b_1	b_2	$\dots$	b_{n-1}	
4	b_{n-1}	b_{n-2}	$\dots$	$\dots$	b_0	
5	c_0	c_1	c_2	$\dots$		
6	c_{n-2}	c_{n-3}	c_{n-4}	$\dots$		
7						

$(2n-3)$ $n \rightarrow$ order power

$$b_0 = \begin{vmatrix} a_0 & a_n \\ a_n & a_0 \end{vmatrix}$$

$$b_1 = \begin{vmatrix} a_0 & a_{n-1} \\ a_n & a_1 \end{vmatrix}$$

$$b_2 = \begin{vmatrix} a_0 & a_{n-2} \\ a_n & a_2 \end{vmatrix}$$

Jury's stability

z-domain

stable - all the poles
should lie inside
unit circle.

R-H

s-domain

stable - left half of
s-plane.

Conditions $|a_0| < |a_n|$

$$|b_0| > |b_{n-1}|$$

$$|c_0| > |c_{n-2}|$$

If necessary and sufficient cond, it means
all the poles of characteristic equation lie
inside the unit circle, $\therefore$ system is stable.

→ Check the stability of sampled data control s/m using Jury's stability test.

$$5z^2 - 2z + 2 = 0.$$

$$F(z) = 5z^2 - 2z + 2$$

A: Chk Necessary cond.

$$\text{Sub } z=1 \Rightarrow F(1) = 5 - 2 + 2 = 5 > 0 \checkmark$$

$$(-1)^n F(-1)$$

$$F(-1) = 5 - 2 \times -1 + 2 = 9$$

$$n=2 \rightarrow \text{power of } z.$$

$$(-1)^2 F(-1) = 9 > 0 \checkmark$$

Necessary cond. are satisfied

Chk sufficient cond.

$$\frac{2n-3}{4-3} = 1$$

Row	z^0	z^1	z^2	
	a_0		a_n	
1	2	-2	5	→ coeff.

$$|a_0| < |a_n| \Rightarrow 2 < 5 \rightarrow \text{satisfied}$$

∴ Suff. cond. are satisfied.

Both necessary and sufficient cond. are satisfied
thus all poles lie inside unit circle, thus
the s/m is stable

→ Check the stability using Jury's stability test

$$z^3 - 0.2z^2 - 0.25z + 0.05 = 0$$

$$A: F(z) = z^3 - 0.2z^2 - 0.25z + 0.05 = 0$$

$$\begin{aligned} F(1) &= 1 - 0.2 - 0.25 + 0.05 \\ &= 1.05 - 0.45 = 0.6 > 0 \quad \checkmark \end{aligned}$$

$$\begin{aligned} F(-1) &= -1 - 0.2 + 0.25 + 0.05 \\ &= -1.2 + 0.3 = -0.9 \end{aligned}$$

$$(-1)^3 F(-1) = -1 \times -0.9 = 0.9 > 0 \quad \checkmark$$

Necessary cond. are satisfied.

Row	z^0	z^1	z^2	z^3
1	0.05	-0.25	-0.2	1
$2n-3$ 2	1	-0.2	-0.25	0.05
$2n-3$ $6-3$ <u>3 rows</u> 3	-0.9975	0.1875	0.24	

$$b_0 = \begin{vmatrix} 0.05 & 1 \\ 1 & 0.05 \end{vmatrix} = \underline{\underline{-0.9975}}$$

$$b_1 = \begin{vmatrix} 0.05 & -0.2 \\ 1 & -0.25 \end{vmatrix} = \underline{\underline{0.1875}}$$

$$b_2 = \begin{vmatrix} 0.05 & -0.25 \\ 1 & -0.2 \end{vmatrix} = \underline{\underline{0.24}}$$

Sufficient cond $|a_0| < |a_n|$

$$0.05 < 1 \quad \checkmark$$

$$|b_0| > |b_{n-1}| \quad |-0.9975| > |0.24|$$

$$0.9975 > 0.24 \quad \checkmark$$

All sufficient cond. are satisfied.

$\therefore$ System is stable or all the poles lie ~~on~~ inside the unit circle.

$$\rightarrow Z^4 - 1.7Z^3 + 1.04Z^2 - 0.268Z + 0.024 = 0$$

Check stability using Jury's stability test.

$$A: F(Z) = Z^4 - 1.7Z^3 + 1.04Z^2 - 0.268Z + 0.024 = 0$$

$$F(1) = 1 - 1.7 + 1.04 - 0.268 + 0.024$$

$$= 0.096 > 0 \quad \checkmark$$

$$F(-1) = 1 + 1.7 + 1.04 + 0.268 + 0.024$$

$$= +4.032$$

$$(-1)^4 F(-1) = 4.032 > 0$$

Necessary cond are satisfied.

Row = $2 \times 4 - 3 = 8 - 3 = 5$

Row	z^0	z^1	z^2	z^3	z^4
1	0.024	-0.268	1.04	-1.7	1
2	1	-1.7	0.024 1.04	-0.268	0.024
3	-0.994	1.6936	-1.015	0.2272	
4	0.2272	-1.015	1.6936	-0.994	
5	0.9472	-1.462	0.6296		

Ktunotes.in

$$b_0 = \begin{vmatrix} 0.024 & 1 \\ 1 & 0.024 \end{vmatrix} = -0.994$$

$$b_1 = \begin{vmatrix} 0.024 & -1.7 \\ 1 & -0.268 \end{vmatrix} = 1.6936$$

$$b_2 = \begin{vmatrix} 0.024 & 1.04 \\ 1 & 1.04 \end{vmatrix} = -1.015$$

$$b_3 = \begin{vmatrix} 0.024 & -0.268 \\ 1 & -1.7 \end{vmatrix} = 0.2272.$$

$$c_0 = \begin{vmatrix} 0.994 & 0.2272 \\ 0.2272 & -0.994 \end{vmatrix} = 0.9472$$

$$c_1 = \begin{vmatrix} -0.994 & -1.015 \\ 0.2272 & 1.6936 \end{vmatrix} = -1.462$$

$$c_2 = \begin{vmatrix} -0.994 & 1.6936 \\ 0.2272 & -1.015 \end{vmatrix} = 0.6296$$

Sufficient cond.

$$|a_0| < |a_n|$$

$$0.024 < 1 \quad \checkmark$$

$$|b_0| > |b_{n-1}|$$

$$0.994 > 0.2272 \quad \checkmark$$

$$|c_0| > |c_{n-2}|$$

$$0.9472 > 0.6296 \quad \checkmark$$

All suff. cond. satisfied

Both necessary &
suff. cond satisfied
 $\therefore$ stable s/m

Stability Analysis using Bilinear Transformation

The transformation is performed by substituting

$$Z = \frac{1+z}{1-z} \text{ in the characteristic eq. of the s/p.}$$

Charac. eq.

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0$$

Sub $z = \frac{1+z}{1-z}$

$$a_n \left(\frac{1+z}{1-z} \right)^n + a_{n-1} \left(\frac{1+z}{1-z} \right)^{n-1} + \dots + a_1 \left(\frac{1+z}{1-z} \right) + a_0 = 0.$$

Eq. can be simplified into

$$b_n z^n + b_{n-1} z^{n-1} + \dots + b_1 z + b_0 = 0.$$

Necessary cond $\rightarrow$ all the coeff. are +ve.

Sufficient cond $\rightarrow$ all the elements in
first column of Routh
array is +ve or no sign
change.

The system is stable if both necessary & sufficient cond. for stability are satisfied.

→ Check for stability of the sampled data control s/m represented by the following characteristic equation.

$$Z^3 - 0.2Z^2 - 0.25Z + 0.05$$

A: $F(Z) = Z^3 - 0.2Z^2 - 0.25Z + 0.05$

Sub $Z = \frac{1+r}{1-r}$

$$F(Z) = \left(\frac{1+r}{1-r}\right)^3 - 0.2 \left(\frac{1+r}{1-r}\right)^2 - 0.25 \left(\frac{1+r}{1-r}\right) + 0.05 = 0$$

Multiply by $(1-r)^3$

$$(1+r)^3 - 0.2(1+r)^2(1-r) - 0.25(1+r)(1-r)^2 + 0.05 = 0$$
$$(1+r)(1+r)^2 - 0.2(1+r)(1+r)(1-r) - 0.25(1+r)(1-r)(1-r) + 0.05 = 0$$

$$(1+x)(1+2x+x^2) - 0.2[(1-x^2)(1+x)] - 0.25(1-x^2)(1-x) + 0.05 = 0$$

$$1+2x+x^2+x+2x^2+x^3 - 0.2[1+x-x^2-x^3] - 0.25[1-x-x^2+x^3] + 0.05 = 0$$

$$1+3x+3x^2+x^3 - 0.2 - 0.2x + 0.2x^2 + 0.2x^3 - \cancel{0.25} + 0.25x + 0.25x^2 - 0.25x^3 + 0.05 = 0$$

$$(1-0.2-0.25+0.05) + x(3-0.2+0.25) + x^2(3+0.2+0.25) + x^3(1+0.2-0.25) = 0$$

$$0.6 + 3.05x + 3.45x^2 + 0.95x^3 = 0.$$

$$0.95x^3 + 3.45x^2 + 3.05x + 0.6 = 0$$

Necessary condition $\rightarrow$ all the coefficients are +ve.

Necessary cond. is satisfied.

Sufficient cond. can be determined by constructing Routh array.

$$s^3 \left| \begin{array}{c} 0.95 \\ 3.05 \end{array} \right|$$

$$s^2 \left| \begin{array}{c} 3.45 \\ 0.6 \end{array} \right|$$

$$s^1 \left| \begin{array}{c} 2.88 \end{array} \right|$$

$$s^0 \left| \begin{array}{c} 0.6 \end{array} \right|$$

$$\frac{3.45 \times 3.05 - 0.95 \times 0.6}{3.45} = 2.88$$

$$\frac{2.88 \times 0.6 - 3.45 \times 0}{2.88}$$

There is no sign change in first row, i.e. all poles lie on left half of s-plane.

$\therefore$ Sufficient condition is satisfied.

Both necessary and sufficient conditions are satisfied. $\therefore$ Stable system.